BRIAN LIM WEI SHENG

AI | Backend | Data Engineer

KL, Malaysia ⋄ Open to Relocate ⋄ Remote ⋄ Hybrid ⋄ On-site

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PROFILE

Education BSc (Hons) Computer Science with Artificial Intelligence (GPA: 3.3) *2020 –- 2023*
Key Courses: Machine Learning, Artificial Intelligence Methods, Algorithms Correctness & Efficiency, Database and Interfaces, and Operating Systems & Concurrency

Summary Computer Science graduate from the University of Nottingham and a full-time Data Engineer with hands-on experience building end-to-end cloud data pipelines for analytics, compliance, and reporting. Proficient in Python and SQL, with practical expertise in ETL/ELT pipeline development, API data ingestion, and cloud data platforms such as Google BigQuery. Experienced in transforming large-scale raw data into structured, analytics-ready datasets.

SKILLS

Languages/Tools: Python, SQL, Java, HTML/CSS, TensorFlow, Keras, Pandas, Git, PostgreSQL, Google Cloud, MySQL, Jupyter, Tableau

EXPERIENCE

Data Engineer, DataQ Tech *Mar 2025 - Present*

- Designed and maintained end-to-end Pentaho/BigQuery data pipelines processing up to 1 million gaming records per month, delivering analytics datasets for 10+ product.
- Built robust vendor API ingestion flows for 3 external data sources; normalized payloads and loaded to BigQuery with < 2% data loss, supporting daily executive dashboards and ad-hoc analysis.
- Achieved 95% SLA success for ingestion job completion (off-peak season), reducing latency for downstream reporting and improving Ops decision-making.
- Implemented withdrawal reputation checks that held >85% of suspicious payout requests before approval.

Skills: Python, PostgreSQL, MSSQL, ETL, REST APIs, Prompt Engineering, Cursor, Postman, Azure Repo, Git, Cronjob, API Authentication, API Integration, Linux CLI, Agile Development

SABA Intelligence Data DLT

- Built a secure vendor API ingestion pipeline to Google BigQuery with encrypted credential handling; implemented Dataform to enforce Type-2 SCD architecture, enabling historical tracking of player health metrics for fraud detection and risk mitigation.
- Designed a DLT pipeline with backfill and configuration modification capabilities, eliminating intermediate Postgres staging datasets and reducing end-to-end pipeline latency.
- Enabled real-time fraud detection and audit capabilities while significantly reducing data pipeline latency through optimized DLT architecture.

Skills: Python, Dataform, DLT, Tableau, Postman, OAuth 2.0, Confluence, REST APIs, Linux, Shell Scripting, Bash

Pentaho Job Alerting System

- Developed a real-time monitoring system that tracks Pentaho job logging files' last modified timestamps and sends automated Teams alerts when job completion times exceed configured thresholds.
- Improved incident response times for the on-call support team by enabling proactive detection and alerting of failed or stalled data ingestion jobs.

Skills: Python, Pentaho, Linux, Shell Scripting, Bash, Microsoft Teams

Digital Audit Internship, PwC Malaysia *Oct 2024 –- Jan 2025*

- Assisted banking clients in identifying and mitigating risks associated with their data systems, improving data integrity and compliance with regulatory standards.

- Utilized SQL to extract and validate 10,000+ rows of financial data related to credit cards and housing loans, ensuring data reliability.
- Conducted 4 IT General Controls (ITGC) tests, ensuring compliance with industry standards and verifying data accuracy in financial reports.

Skills: SQL, Microsoft Excel, Risk Assurance, ITGC Controls

Data Engineering Internship, Gamers2Gamers SDN BHD *Jun 2022 – Aug 2022*

- Developed a Google OCR model integrated into the company's P2P gaming platform registration portal, enabling automated text extraction from user-submitted ID documents for streamlined account creation.
- Manually annotated 10,000+ images of IDs and passports, labeling key fields such as photos, addresses, ages, and genders to train the OCR model for accurate text recognition and machine-readable data transformation.
- Configured Python scripts to optimize OCR performance across diverse document layouts (e.g., varying formats for US vs. Canada driver licenses), ensuring reliable extraction and integration into the registration workflow.

Skills: Python, Google OCR, Machine Learning

PROJECTS

House Price Regression Problem *2024*

- Built an ML pipeline for the Ames housing price dataset (2,919 rows, 81 features) to predict SalePrice from location and property features (rooms, area, quality, etc.), enabling data-driven valuation insights.
- Evaluated ElasticNet, Lasso, Random Forest, and XGBoost models; applied log transform to SalePrice to normalize skew, stabilize variance, and improve regression performance.
- Selected ElasticNet as the top model with RMSE 0.11657 (log-scale) for its balance of regularization and correlated-feature handling, delivering robust generalization on structured housing data.

Skills: Python, scikit-learn, Pandas, TensorFlow, Keras, Regression, RMSE evaluation

Credit Card Clients Clustering Analysis *2024*

- Performed credit-risk segmentation on client data using unsupervised clustering to identify creditworthiness groups and support high-risk applicant rejection decisions.
- Used Elbow Method with WCSS to determine optimal cluster count (k=3), improving segment stability and business interpretability for credit policy enforcement.
- Derived 3 actionable clusters (highly active applicants, inactive applicants, payment-behavior segments such as cash-in-advance only vs installment-only) to inform targeted risk-based lending strategies.

Skills: Python, scikit-learn, KMeans, PCA, data preprocessing, EDA, WCSS, clustering

Final Year Project – Heart Disease Prediction Classification *2023*

- Built a Cleveland dataset classification pipeline using clinical features (age, sex, resting blood pressure, cholesterol, max heart rate, chest pain type) for heart disease detection.
- Used SelectKBest and RFE to identify core predictors in human-readable terms: age, max heart rate, chest pain type, exercise-induced angina, ST depression (oldpeak), and calcium channel counts (ca/thals).
- Benchmarked Logistic Regression, KNN, SVM, Decision Tree, Random Forest, MLP, and XGBoost using accuracy, AUC-ROC, and confusion matrix metrics.
- Performed hyperparameter tuning (GridSearchCV + K-fold CV), improving test accuracy from 78% to 83% and highest AUC to 0.892 for XGBoost.

Skills: Python, scikit-learn, pandas, NumPy, Matplotlib, GridSearchCV, K-fold CV, ROC/AUC